Data administration

PID_00302162

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Recommended minimum reading time: 3 hours

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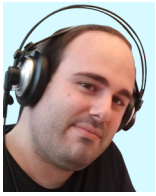

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The assignment and creation of this UOC Learning Resource have been coordinated by the lecturer: David Bañeres Besora

How to cite this learning resource as per the Harvard style:
Mendoza Flores, M., Colobran Huguet, M., and Panadero Martínez, J. (2026) *Data administration* [Textual learning resource]. 1st ed. Barcelona: Fundació Universitat Oberta de Catalunya (FUOC).

First edition: February 2026
© of this edition, Fundació Universitat Oberta de Catalunya (FUOC)
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Introduction

Information (also known as data, although we will see the difference between the two concepts later) **is currently one of the most valuable assets in any organisation**. This is the raw material that feeds the IT system. Therefore, it is crucial to know where it is located because all computers are connected via IT networks, which means that data is at risk of being scattered across all the organisation's workstations. If this is the case, the organisation will not know what data it has or where to find it.

The organisation need to know where the information is stored for backup purposes. Although the software can be reinstalled in the event of a disaster, the data was created by the organisation and cannot be recovered without a copy. Since information is one of the most important assets today, it must be protected.

The location of this data is also important so that it can be combined to obtain new insights about an organisation.

For the organisation management (and the organisation as a whole), access to the right information at the right time leads to better decision-making and timely action.

Objectives

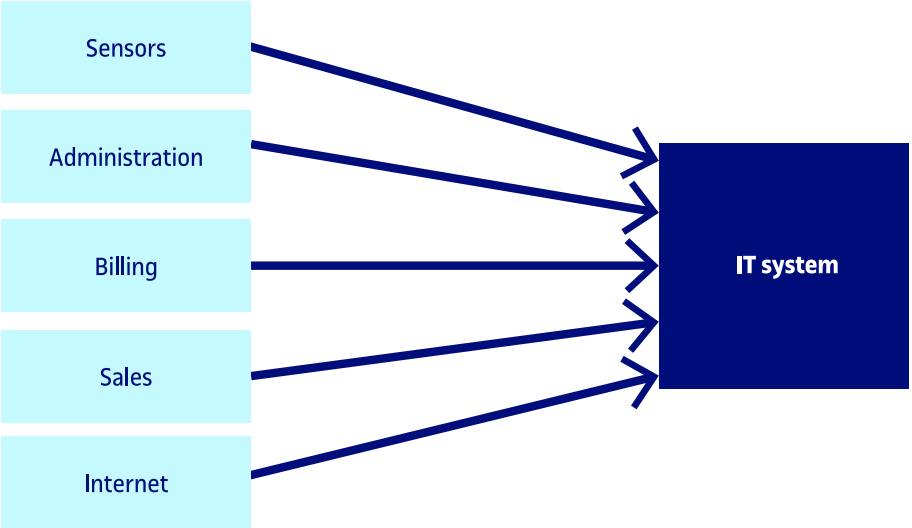
The learning resources in this module cover the content and key tools needed to achieve the following objectives:

1. To understand the high value of organisational information.
2. To know how important it is to determine where all organisational information is located.
3. To be familiar with different ways of maintaining information integrity.
4. To learn why database servers and other data architectures have become popular.
5. To know where organisational data may be found and where it should ideally be stored.
6. To be familiar with database security concepts.

1. Data and organisations

Organisations are constantly generating data, so IT systems can be viewed as simply data warehouses. Figure 1 shows the diversity of data sources.

Figure 1. An IT system can collect data from various sources



However, this extensive repository of raw data lacks practical utility in its unprocessed form, as its sheer volume renders it ineffective without proper organisation and standardisation.

Before moving on, the difference between information and data must be explained. The formal definitions of these two concepts are as follows:

Data is a record of events.

Information is data that has been processed to make it meaningful.

For instance, the number 70293 could represent either 7 February 1993 or package 7029 for lorry number 3. Similarly, 1813 could be a car number plate or the time 18:13.

Data and information

Despite this formal difference, these two words are often used interchangeably by those working in IT. Since processing data to make sense of it can be done in many places, the terms *data* and *information* can be used interchangeably.

Data is generally stored and shown to the user, i.e. the processing of this data so that it makes sense is carried out in the software when it is displayed on the output device. Fundamentally, this gives the system greater flexibility, as it allows data to be converted to any output format. However, this is often far from reality because different software programs store data in different formats. Furthermore, most software programs are incompatible with each other, leading to difficult cross-platform data recovery.

As a result, organisations constantly create large amounts of data, which computer systems process and distribute among all the components of the organisation to increase overall efficiency.

> **IT systems** aim to store the organisation's information so that it can be easily retrieved later in the same way it was stored.

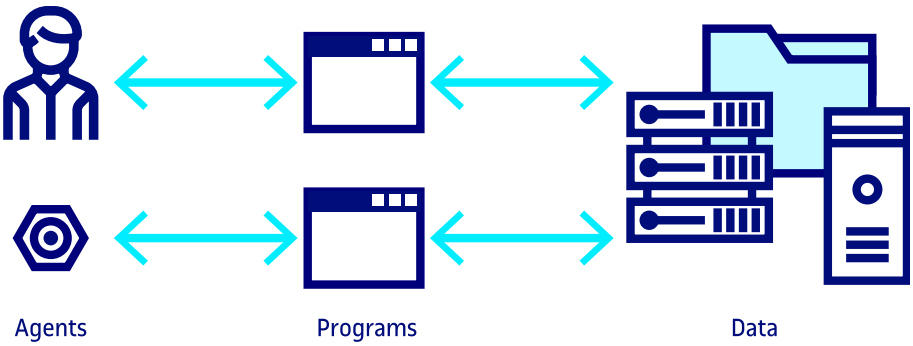
The key objective of IT systems is to store information and make possible to retrieve it in the same way it was stored. Nevertheless, this negates one of the computer's features which is their capacity to handling information.

Here, the term "handling information" is quite broad. Below are some possible meanings:

- Adding new information to what already exists.
- Deleting obsolete information.
- Modifying incorrect information.
- Linking different pieces of information to construct new information, checking the consistency or accuracy of the information to be added, reducing information redundancy or finding new relationships.

The interaction between data and the organisation can be considered as follows:

Figure 2. Interaction chain between users, programs and data



As shown in Figure 2, the data interaction chain has three main components: agents, programs and data. Each of these is defined below.

- **Agents**. They can be people or sensors that access the organisation's data via programs. These agents could also be considered as entities outside the organisation on the internet that interact with the IT system and modify its data.

- **Programs**. They are used to access the organisation's data orderly and properly. Given that data is a valuable organisational asset, it seems logical that access to it should be protected. Together with the operating system, programs are responsible for carrying out all the necessary controls to prevent unwanted access to the organisation's data.

The task of extracting specific information from a computer system can appear straightforward and feasible. Based on our observations so far, this issue would be solved by creating a new program that accesses the data. However, this does not address the issue of management of permissions and group access for various files to the organisation's information. Instead, it is essential to safeguard sensitive content against unrestricted access.

The IT system collects, stores, handles and presents information to users.

2. Where is the information?

Unfortunately, it is not so straightforward in the real world. In the previous section, everything was ideal. The data was stored and, as needed, it could be retrieved and presented. If the end goal was reached, the job was done. The reality is that many more processes are involved that further complicate the system.

Here are some of the challenges:

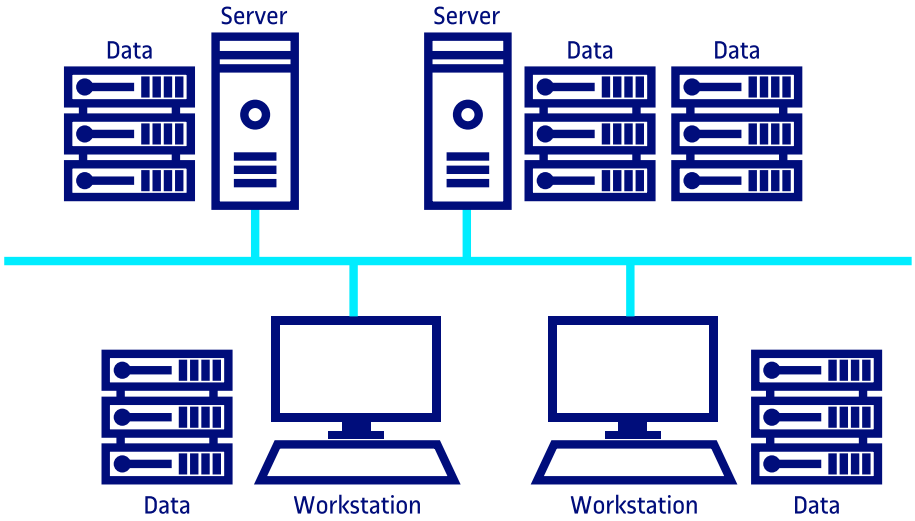
- Data is stored in files.
- Data is stored in a certain format.
- Data is not always stored where it is needed.
- Data sources, i.e. sensors, people, web forms, etc., generate data in a certain format.
- Each data source may use a different format and may be stored in potentially different files and locations.

Different web forms may generate different text files. Depending on the application used, a single user will create different information in different formats that will end up in separate files. For instance, when preparing word processor documents for the bank's annual balance sheet in separate files according to different accounts, related information on the same subject may be generated in multiple locations, which can make linking the documents challenging.

Therefore, in practice, the data of the organisation will be scattered at the software level. Not all programs will be able to access all data, regardless of the user's permissions.

The issue becomes even more problematic when looking at the structure of the network. Data may be in one place because it has been chosen that way or because users, unknowingly, generate information and store it in that place, without checking it.

Figure 3. Data distribution across multiple sites within the organisation



As shown in Figure 3, information can be stored in any directory on any partition of any disk on servers or workstations, even on laptops.

Extreme cases exist where duplicate information is stored on different machines with different operating systems and different file systems.

To make matters worse, some of this information may be critical and its location may be unknown by the IT department, or perhaps they know it exists but have no control over it and therefore have no disaster recovery mechanism in place.

2.1. Possible solutions

As the previous section has stated, this is a potentially serious issue that has been historically prevalent in all organisations. The following is a summary of some possible solutions.

1) Based on the application chart and the general user design, it is possible to determine which software users have access to. The application chart also provides information on the location of the information. The user administrator or server administrator is responsible for maintaining the information stored on the servers. Regardless, the location of the information is clear, and appropriate security, protection, and recovery mechanisms are in place for emergencies.

See also

See how this is done in the *User administration* module.

Based on the application chart and the general user design, it is possible to determine which software generates information on the local system. All this software must be backed up using the appropriate mechanism, and the nec-

essary permissions and security measures must be enabled to guarantee confidentiality. It must also be incorporated into the general backup mechanism so that it is protected in the event of failure.

2) "Uncontrollable" information. This category typically includes office software (spreadsheets, word processors, small databases, calendars, etc.). It is important to make sure that the data is stored on the user's private network drive (which should be backed up daily).

3) User training should also be provided on how to use their IT tools so that they learn to save information on their personal network drive rather than on the local drive (or saving it on the application default location, as this often means that even the user does not know where the data has actually been saved). This "IT culture" can prevent information from being scattered across all the organisation's disks, which is a major administrative issue.

Reflection

Do you know where information is stored in your organisation? Please share your thoughts in the course forum.

Mobile computing

With the growth of mobile computing in organisations, dispersion is now compounded by the fact that these devices need to be autonomous and available at all times. This raises the first issue: what happens to the data needed to work? Such data may be sensitive if it leaves the organisation unprotected. The second issue is that these devices modify or add data to the computer system (they are disconnected from the network). How do they synchronise this information with the organisation's system? The final issue is that sometimes they need to operate connected to the organisation's network as if they were just another workstation, and sometimes they need to connect to the organisation's network remotely. How can this be done?

Methods must be identified to determine the location of all the organisation's information. This will be useful to guarantee its security.

3. Information retrieval

While the organisation constantly generates large amounts of data, users also need to access new information within the organisation frequently. These may be new queries to the data or new data sources within the system.

As a result, the administrator must be familiar with all of the organisation's databases.

Databases

The term *database* is used broadly in this context, covering everything from a spreadsheet to a complete database, or even a word processing document. Everything is organisational information.

Since the administrator is the responsible for all organisational information, it is essential to:

- Know where it is located.
- Have a recovery method in case of problems. This involves backups.
- Have a general idea of the content. Have information about the information (we could call this meta-information).

Locating the information is important because it has a major impact on backup policies and access times for users, as well as ease of access, future expansion, expansion of the user group accessing the data, etc.

The general idea of the content is useful to optimise the system (databases and data already included should not be repeated) and to meet management requests, which usually involve extracting information from the system.

Data mining

Data mining methods seek to extract coherent information from large amounts of incoherent and often scattered data.

Even when we manage to put all the data on servers (on some partitions on some disks on some servers), the problem is not solved, but at least it is a big step in the right direction. There are now a few options available:

- Different groups/users can access the same data (share it). This makes it possible to add items and query, modify and delete data.
- Since the data is on the servers, security is much tighter.
  - Backups of the information.
  - Loss due to hardware failure.
  - Theft.
  - Permissions and, therefore, access restrictions.

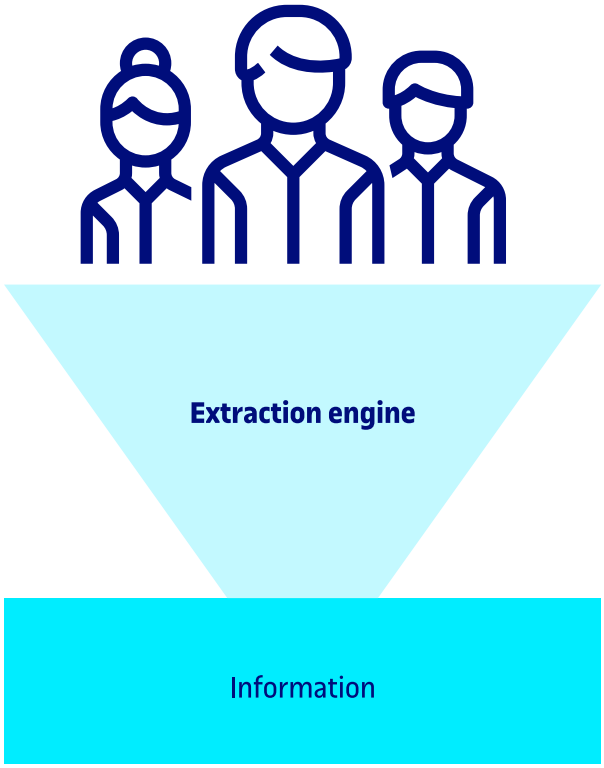
- Easier access and dissemination. Under these conditions, the information is potentially accessible to the entire organisation (it is controlled by the security system).

3.1. Management queries

It is now time to consider an issue that has not been addressed until now: management queries.

Management views the IT system as a data warehouse and queries this information to support decision-making.

Figure 4. Organisations perform various searches using the data



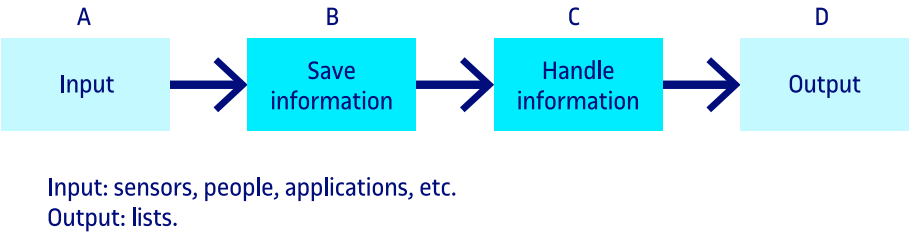
As shown in Figure 4, management sees the IT system as an abstract black box, which it queries when making decisions with questions such as "Which product has generated the fewest production hours and the highest sales profit?", or "Which product has cost the sales team the least to sell?", or "Which products have the lowest transport costs?", or, finally, for example, "Which employees have been the least productive?".

All of these questions are often quite difficult to answer, although the administrator may know that the information is in the system.

Previously, when data was distributed through the organisation, it was impossible to answer any of these questions. Now that data is stored on servers, everything is accessible. **What is the problem?**

To answer this question, it is necessary to take a step back and analyse how the organisation's information has been obtained in the past. Figure 5 shows the process of saving/retrieving information.

Figure 5. Data transformation cycle



So far, only step B of the flowchart (saving information) has been solved, since all the information is accessible within the servers. Alongside this, the security and information sharing issues have been improved, but no progress has been made on step A: information continues to arrive in completely heterogeneous formats, spread across a large number of different files, meaning that the software needed to handle this information can be extremely complex to develop. If handling information in heterogeneous formats to obtain coherent information is complex, and management makes frequent queries, the problem becomes unmanageable.

Finally, the result obtained from some of these files with heterogeneous formats will also be in a heterogeneous format (for example, the same information could be stored in two different ways). How to develop the final list (step D)?

> Simply having all the information on the servers is not enough to claim that the system is ready to process it as desired.

3.2. Database servers

Database servers emerged as a means of efficiently handling all of a corporation's information and making it useful for decision-making.

These servers were developed to meet companies' need to manage large volumes of heterogeneous data while sharing information with a group of people quickly, securely and easily.

The most commonly used data systems today can be reduced to two types, each with different configurations.

1) Relational databases

A relational database is a collection of items with defined relationships between them. These are organised into tables of information where each entry represents a row with different fields or columns. Tables are used to store information about the items they represent in the database. Each column in a table stores a specific type of data and each field stores the value of the attribute. Similarly, each row can contain the index key that should be used to perform searches.

Relational databases have the following features:

- Data integrity, based on the relationships established between different fields.

- ACID compliance. Each transaction must therefore comply with the following criteria:
  - atomicity
  - consistency
  - isolation
  - durability

- This can be modelled as a transactional system, where each request is independent of the previous one and forms a single logical work unit.

- Searches can be performed using SQL.

2) Non-relational or non-SQL databases

Non-SQL databases are data structures that do not support the SQL query language and do not comply with the ACID properties mentioned above. They appeared at the same time as big data, driven by the need to store and manage large volumes of data efficiently. Typically, these non-SQL databases use other data structures that are easier to scale.

There are several types of databases depending on the type of data:

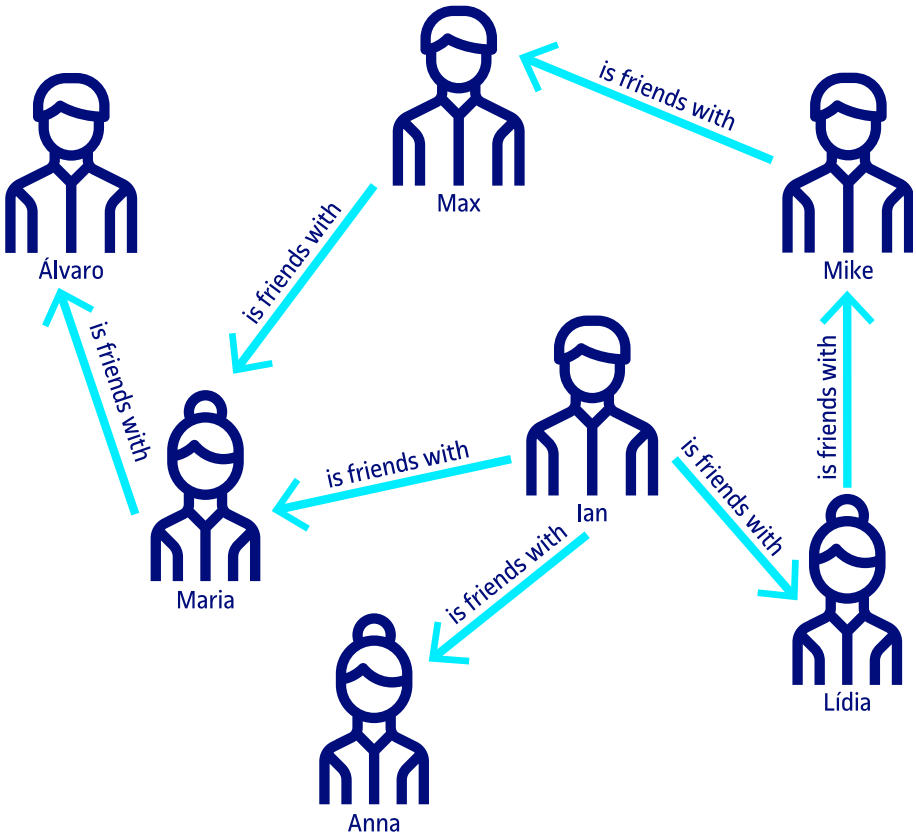
a) Key-value. A key-value database is a simple database that uses an associative arrangement as a data model, where each key is associated with a single value in a collection. This relationship is known as a key-value pair. This is a straightforward method of storing data, which is easy to scale. The data in this database is highly divisible and can be easily scaled horizontally.

b) Document. Document databases are often the most efficient way to store and index information. This type of database has been designed to store and query data such as JSON documents. The flexible, semi-structured and hierarchical nature of documents, and the data they contain, means that they can

be adapted to the needs of the organisation. The document model works well with use cases such as catalogues, user profiles and content management systems, where each document is unique and changes over time.

c) Graph. This is used to create databases with highly interconnected data. Graphs are really useful for capturing complex relationships in big information networks. They are shown as a graph where the nodes and edges hold the query information. The aim is to display information in a visual way so that it can be read more naturally. Figure 6 shows an example of the result of a query to a graph database, where the relationships between a set of people are represented.

Figure 6. Example of a query result in a graph database



3.3. Large-scale data processing

The following section describes the different techniques used to work with large volumes of data.

3.3.1. Big data

Big data can be defined as large-scale data analysis from various sources whose volume is beyond the capacity of traditional data processing software applications to gather, process and interpret within a reasonable time frame. This

phenomenon has led to the emergence of new technologies that make it possible to store and process the information obtained through these technologies, as well as use it.

There are usually several sources of data that can generate these volumes:

- People who use large-scale IT resources such as Meta, X, etc.
- System communications, such as logs, M2M data, IoT devices that send telemetry, etc.
- Marketing data, where increasing amounts of user data are being collected.

The volume of data managed today is much higher than a few years ago and that it will continue to grow. The following data properties are worth highlighting:

1) Speed: nowadays, information comes from different sources and needs to be processed and stored immediately. The concept of immediacy has changed in recent times, and information is sought almost instantly. As a result, the speed of analysis required by today's society is one of the fundamental features of big data, where information is constantly moving and processed in real time, executing increasingly complex algorithms in less time.

2) Variety: depending on where it comes from, the volume and frequency of this data can be totally different. For example, tweets are really small, but on the other end of the spectrum, high-resolution videos take up a lot of storage space in comparison. The amount of unstructured data is also growing faster than traditional data. Like volume, unstructured data requires a new approach to information processing, new methodologies and technologies for analysis.

3) Value: this is a key factor when managing the data life cycle. Value is undeniably a core quality in analysis and one that can have the most direct influence on how data is handled.

4) Variability: with big data, things change fast, so information varies a lot, which means that the models or processes used to handle it need to be checked regularly.

5) Volume: as previously mentioned, the quantity of data generated is increasing. As databases grow, applications and the architecture built to support the collection and storage must also grow. Although the price per TB of storage has fallen considerably, cost remains one of the key factors in correctly scaling a system.

6) Accuracy: the information collected must be accurate. This is crucial for obtaining high-quality data and is essential for careful analysis, depending on the intended use of the data.

7) Visualisation: presenting data in an easily understandable format is essential to provide value to the business.

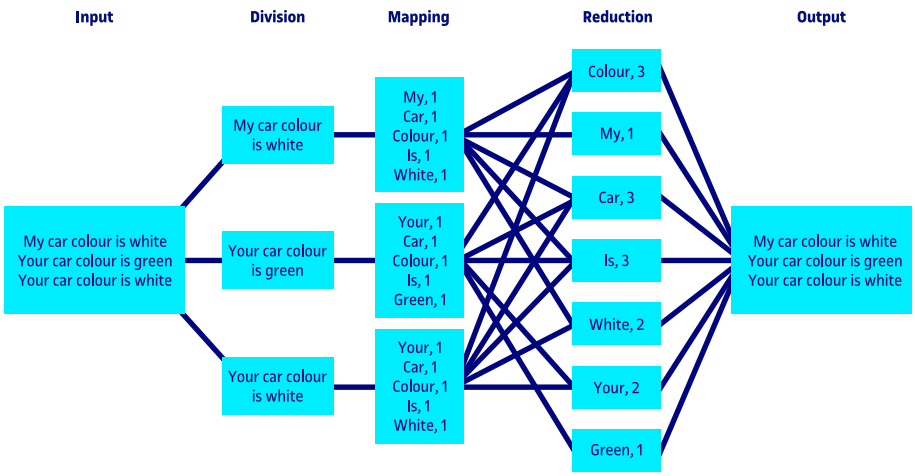
Working with such large volumes of data requires efficient processing and storage techniques. Below is a widely used technique.

3.3.2. Map reduce model

The advent of big data has led to the development of techniques to process such large volumes of data. One of these techniques is map reduce, which involves an initial process of mapping, filtering and sorting the data, typically on parallel servers. Once the data has been formatted, it is then reduced.

Figure 7 shows an example of a reduced map. As shown, there is an input dataset which, after being divided and mapped (key, value), can be sorted by key (in the example below, the key corresponds to the names, together with the values of each input). Next, the reduction step groups the different keys by adding the values, resulting in an efficient indexing of the content.

Figure 7. Map reduce example, showing the different stages

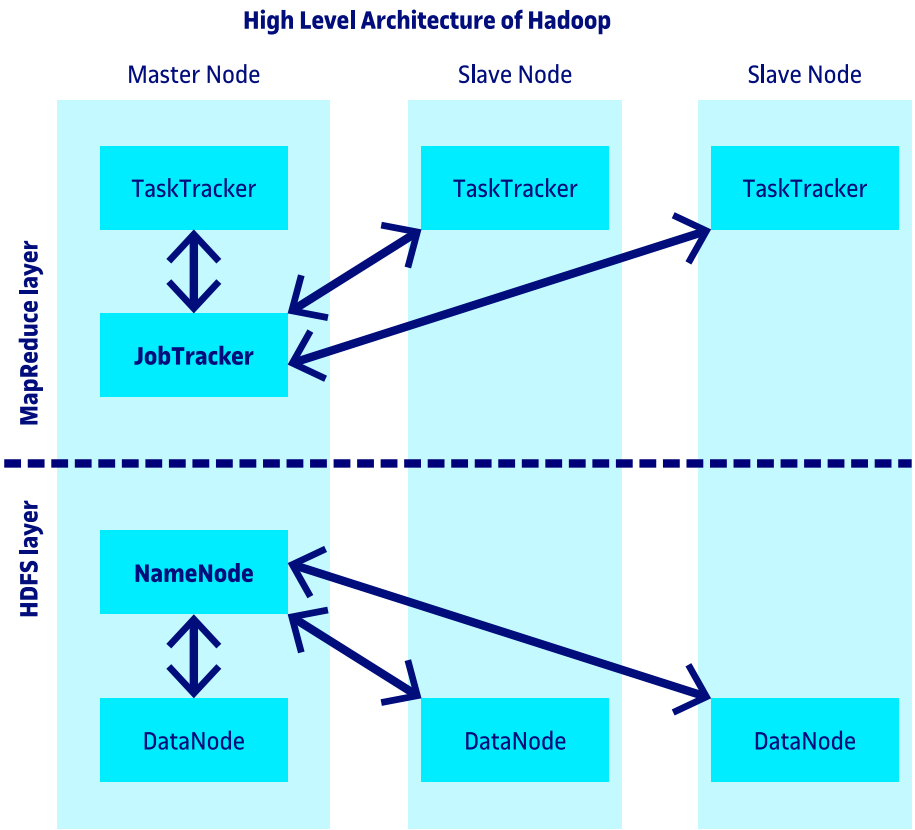


1) Hadoop

Hadoop is an open source framework for storing, processing and analysing big data, making it easier for developers to deal with parallel programming.

To store data, Hadoop has created HDFS, Hadoop Distributed File System, written in Java, which distributes data across different nodes with multiple replicas to ensure high availability. As shown in Figure 8, the data is distributed and available across the different nodes that make up the HDFS cluster.

Figure 8. Hadoop architecture with HDFS



Source: adapted from https://upload.wikimedia.org/wikipedia/commons/8/85/Hadoop-HighLevel_hadoop_architecture-640x460.png.

To process data, Hadoop uses the map reduce system, which is typically combined with the HDFS data system.

Once data is input, it is first separated so it can be stored in different cluster members and parallel computing algorithms can be run on it.

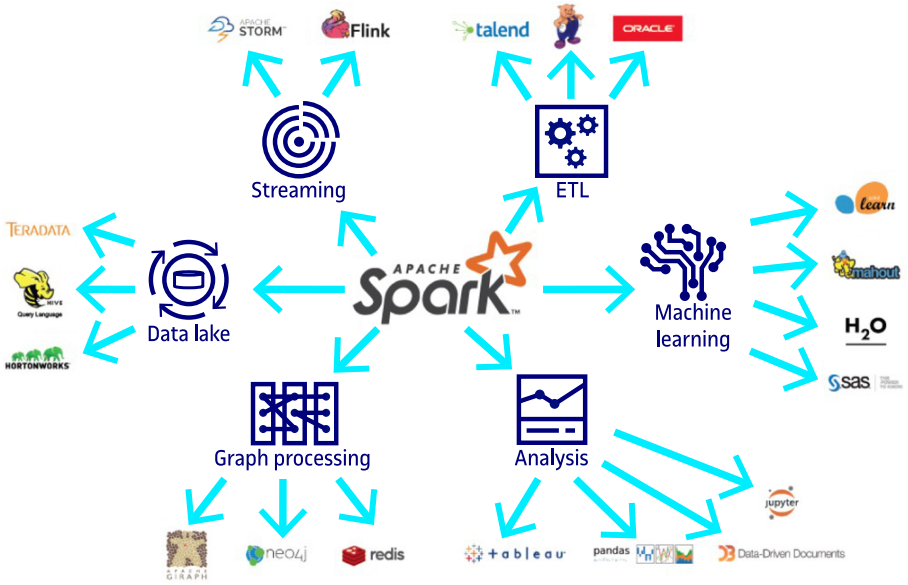
Once the information has been split up, key-value mapping is performed. The data can then be processed, combined or sorted, and finally reduced and presented.

2) Spark

Spark is an alternative option that focuses more on big data processing. Unlike Hadoop, it processes data much faster, primarily because it works with data in memory, thereby reducing slower read/write operations on the disk.

In contrast to Hadoop, Spark can read data from different file types (HDFS, Amazon S3, Google Cloud, etc.), with a strong focus on fast data processing.

Figure 9. Spark makes it possible to work with different types of data and connectors



As shown in Figure 9, Spark offers greater flexibility when performing heavy workloads, featuring iterative algorithms, streaming processing, Python APIs, etc. To do this, Spark has many connectors and provides support for different types of data repositories and analyses.

3.4. ERP

The concept of database servers is now much more advanced. The applications used by all organisations are installed on a database server. The product is then put on the market.

The result is called *ERP* or enterprise resource planning. It is a set of modules or packages (applications), such as accounting, billing and payroll, which are all perfectly integrated and running on a database server.

By integrating and automating many operational aspects of the organisation, these information management systems streamline the entire process of data input and reporting. This integration of all information into a single location makes it possible to perform many powerful queries.

3.5. Cloud data management

Sending data to the cloud is becoming increasingly common. Therefore, several factors must be considered to facilitate migration management and, in particular, ensure the security of the migrated data.

1) Data security. There are two variants:

a) Data security in transit, while the data is being sent or queried. It is always recommended to use secure protocols, such as HTTPS.

b) Security in permanent storage, which involves managing encryption keys. There are two options:

- Encryption key provided by the customer.
- Encryption key managed and provided by the service provider.

2) Data access management. Sharing files is easier with the cloud, although it should never be forgotten that each file has access permissions that must be respected even when accessed from the cloud.

3) Backups and disaster recovery. Although the cloud offers more backup options, copies and continuity mechanisms are key to good data governance.

4) Data governance and quality. While file sharing in the cloud is easier, remember that information must be consistent and have a single source.

Organisations can also implement data leak prevention (DLP) systems, where each file is classified and protected individually to prevent unauthorised access. Microsoft Information Protection, which is integrated with the MS Office 365 suite, is a good example of this.

3.6. Remote repositories

The cloud now offers various ways to store information online. The following is an analysis of some of the systems currently available.

- Storage of items, such as office files.

- Storage of files in NAS format, which serve as a file repository. Examples include Amazon Elastic File System and Azure NetApp Files.

- Block Storage. Widely used by database or ERP systems.

4. Information protection

Information protection can be approached from two angles. The first is simply protecting information from unwanted access. The second is protecting information from loss.

Information is one of an organisation's most valuable assets. A significant portion of efforts must be invested in protecting it.

4.1. Network security

Information is stored on the organisation's network on servers, workstations, laptops, etc. Employees access this information through access mechanisms that limit and control who can access it, how they can access it and what they can do with it. A well-designed user environment is the first step in preventing unauthorised access to information.

Another step towards securing systems is to prevent intrusions. It is very challenging to have completely secure systems, so the best approach is to make the system as secure as possible.

4.2. Backups

Data backups in an organisation have become one of the most important issues in data management. Without a good disaster recovery policy, information can be lost. The basic rule is that data should be backed up according to the backup policy established by the organisation and that a backup history should be kept, which, depending on the organisation, can range from a few months to years.

Backup policies are key to a company's continuity in the event of an IT incident. Therefore, the following points should be reviewed as part of the backup policy.

1) Check an information inventory.

2) Control access to information, making it accessible only to authorised personnel.

3) Validate critical information backups.

4) Backup frequency. Several aspects must be considered:

- Variation in data sources.
- Cost of storing backups.
- Regulatory compliance needs to be fulfilled.
- Types of backups:
  - **Complete**: a backup of all data.
  - **Incremental**: only data that has changed since the last backup.
  - **Differential**: data that has changed since the last complete backup.

5) Backup expiration.

6) Validate that backups are correct.

7) Encryption of information, including backups.

8) Backup location. These backups must be kept in a safe place.

9) If backups are stored in the cloud, the legal and regulatory aspects associated with data storage must also be validated.

10) Finally, once a backup has been deemed no longer valid, it must be disposed of correctly and securely.

4.3. Database security

A database can be understood as a "set of integrated data, suitable for various users and different uses". Therefore, security issues arise when this data is used simultaneously.

Data must be protected against hardware breakdowns, software malfunctions and human errors (whether intentional or unintentional), which can alter or corrupt the data, rendering the database useless for its intended purpose.

All DBMSs must provide techniques that give users access to only part of the database and not to the rest, which is why DBMSs include an authorisation security subsystem designed to guarantee the security of certain parts of the database against unauthorised access.

Databases contain mechanisms to prevent faults (control subsystem), detect them when they occur (detection subsystem) and correct them after they have been detected (recovery subsystem).

The fundamental aspects of database security are:

- **Confidentiality**: not providing data to unauthorised users. This includes privacy aspects (personal data protection).

- **Accessibility**: the information must be available.

- **Integrity**: ensuring that the data has not been corrupted.

4.3.1. Confidentiality

To simplify administration, DBMSs include the concept of user profiles, roles or groups, which combine a set of privileges, so that users belonging to a group are granted all the privileges of that group.

The access control subsystem is responsible for denying or granting access to users. There can be several types of authorisation in an DBMS:

- **Explicit authorisation**: used in traditional systems. These systems store who can access which items in the database and with what privileges. An access control matrix is usually used.

- **Implicit authorisation**: authorisation for an item can be inferred from other items. For example, if a class can be accessed in an DBMS, all instances of that class can be accessed.

- **Positive authorisation**: where it exists, it indicates the existence of authorisation.

- **Negative authorisation**: the explicit denial of authorisation.

4.3.2. Accessibility

Database systems must ensure that data is available to users when they need it. There are mechanisms in place to recover the database in the event of technical or physical failures that destroy all or part of the data.

DBMS security tools

DBMSs contain their own tools. However, from a security perspective, alternatives such as fault-tolerant servers should be considered.

The basic principle behind database recovery following a fault is physical redundancy. The most common faults are caused by power outages, hardware issues, and storage device (disk) errors.

These situations led to the emergence of the concept of **transactions**. When a fault occurs, there must be a way to ensure that the database remains intact after an update. To achieve this, executable units called *transactions* are creat-

ed. These can be defined as sequences of operations that must be executed atomically. Either all operations in a transaction are performed globally, or none are performed at all.

4.3.3. Integrity

In the context of databases, integrity refers to the correctness, validity, or accuracy of the data in the database. The goal is to protect the database from operations that could lead to data consistency issues. A DBMS integrity subsystem should detect and correct incorrect operations as much as possible.

There are two types of operations that can compromise data integrity: semantically inconsistent operations and interference caused by concurrent access.

1) Semantic integrity. Some operations can compromise the restrictions defined in the database design (e.g., restrictions on domains or attributes). These restrictions can be static (also called state or situation restrictions) or dynamic (called transition restrictions).

2) Operational integrity. In multi-user systems, a concurrency control mechanism is required to maintain database integrity. Otherwise, significant inconsistencies could arise from concurrent access.

Summary

Data forms the basis of the IT system and is also extremely valuable to the organisation. The major risk of data is that it can be stored in multiple locations. Efforts should be made to educate users to store data on servers by default. This ensures data consistency and security.

Additionally, the organisation needs to extract new insights from the information. This can only be achieved by centralising the information in a single application, i.e. a database server.

This solution requires a homogeneous interface to be installed on the database server, along with a data architecture, an ERP or a data warehouse.

As information has become a highly valuable asset, it must be protected. Both backup and data access protection policies are therefore essential.

Activities

1. Think about organisations you know (shops, businesses, banks, etc.) and try to identify their possible data sources.

2. Look around you. The same information is presented in many different formats, for example, the time (digital, analogue, etc.). How would you store this in a database? Find other examples of data in multiple formats.

Self-evaluation

1. List some of the data sources for the following places:

a) A hospital.
b) A nightclub.

2. One of the organisation's sales representatives tells you that they don't want to use a database to calculate product sales or to store products and price increases because they have never worked that way. They negotiate directly with customers, and at most they might enter the data into a spreadsheet on their laptop. What would you tell them?

3. Would you centralise the data on servers?

a) Yes, because that makes it easier to share.
b) No, it is more practical to distribute it to the places where it is needed.
c) No, because it could overload the servers and the network.
d) Yes, because it makes it easier to query the organisation's information.
e) Yes, because it completely eliminates redundancy.
f) a and d.

4. Which of the following statements about database servers is false?

a) Database servers concentrate scattered information in a single location.
b) One of the building blocks of ERP systems is a database server.
c) A database server containing the organisation's information makes it easier for management to run queries.
d) A database server reduces network traffic because the output formats are standardised.
e) A database server standardises the way information is stored.

5. One of the following tasks does not fall under the responsibility of the data administrator.

a) Avoiding duplication of information within the organisation as much as possible.
b) Ensuring data availability to users.
c) Configuring corporate databases.
d) Connecting databases to the web server.
e) Ensuring the smooth running of databases.

Answer key

Self-evaluation

1. a) Information is generated in many places. Some of these are:

Patient admission.
Writing prescriptions.
Visits to all clinics.
The entire accounting department.
The entire supplier ordering department.
Payroll department.

b) Information is also generated in many places. Some of these are:

Marketing.
Booking entertainment.
Accounting.
Billing.
Payroll.
Supplier orders.
Security.

2. If someone shares this problem with you, you should convince them using arguments such as:

- The organisation works as a unit, so all prices are in one place, in a database on a server.
- The salesperson can take the data and manipulate it, but if they negotiate price changes, they must reflect these in the database so that the organisation knows:
  - Which salesperson is the most competitive?
  - Which salesperson is performing best?
  - Which salesperson is generating the most revenue?

This information can be used to assign financial or other incentives to salespeople with the best sales performance.

If you only have the data locally and you lose your computer, or it is stolen, etc., neither you nor anyone else in the organisation will be able to recover it. This means that the organisation will lose information and you may lose the incentives mentioned above.

The organisation's information belongs to the organisation. Therefore, the organisation establishes the operating guidelines, not the people who handle the data.

3. f

4. d

5. d

Glossary

data A record of events.

data dictionary A collection of definitions and specifications for data categories and their relationships.

data mart A set of organised facts and data for decision-making support based on the needs of a specific unit or department. The data is intended to meet the specific needs of a department and is only meaningful to the staff in that department.

data mining Data analysis to discover unknown relationships, patterns or associations.

data source Any element (whether belonging to the organisation or not) that creates data used by the IT system.

data warehouse A database that stores a large amount of integrated transactional data for use in analysis.

database General term for a place where data is stored.

database management system Software that organises data so that it can be stored and retrieved.
acronym: DBMS

database server A system that stores data and is asked for information or has information entered into it through some mechanism, usually via network requests.

DBMS See **database management system**.

decision support system Automated application system that helps the organisation make decisions through strategic analysis of historical information.

dimension An independent entity within the multidimensional structure of an organisation that acts as a research key (index) or data selection tool.

DSS See **decision support system**.

enterprise resource planning Integrated information management system designed to provide a complete solution to information issues within an organisation.
acronym: ERP

ERP See **Enterprise resource planning**.

information Processing data so that it makes sense.

OLAP (online analytical processing) A set of principles that provide a dimensional working environment for decision support.

OLTP (online transaction processing) A transactional system that maintains the operational data of an organisation.

SQL See **Structured query language**.

structured query language Language used to query database management systems.
acronym: SQL

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